MedAgent Clinical Decision Support

Case ID: case_0fd718b2b4 | **Status:** APPROVE | **Physician:** Dr. Alex Vance | **Date:** 2026-10-16 09:20 UTC

CRITICAL DISCLAIMER: Decision-support draft generated by AI agents. It is not a diagnosis and must be reviewed, corrected and approved by a licensed physician before any clinical use.

Patient Intake & Presentation

Patient: Age: 23, Sex: Male
Notes & History: .

Clinical Summary / Impression

Clinical evaluation indicates features consistent with Macrocytic / Megaloblastic Anemia, characterized pathophysiologically by deficiency of vitamin b12 or folate disrupting purine/thymidylate synthesis and erythroblast nuclear maturation, leading to nuclear-cytoplasmic dyssynchrony and abnormally large erythrocytes. Key supporting clinical indicators include Macrocytic Anaemia, Vitamin Deficiencies, while secondary differential considerations include Cardiomegaly, Congestive Heart Failure / Pulmonary Edema. Concurrently, chest radiography identifies Cardiomegaly (62.2%). Targeted diagnostic workup and guided serial clinical monitoring are recommended to establish definitive management.

Ranked Differential Diagnoses

#	Condition	Likelihood	Key Supporting Findings	Citations
1	**Macrocytic / Megaloblastic Anemia**	**MODERATE**	Macrocytic Anaemia, Vitamin Deficiencies	S4, S5
2	**Cardiomegaly**	**MODERATE**	Cardiomegaly (p=62.2%)	S3, S7
3	**Congestive Heart Failure / Pulmonary Edema**	**LOW**	Cardiomegaly (p=62.2%)	S3, S7

Radiographic Findings (Chest X-Ray)

Pathology / Finding	Probability	Location	Indicative Severity	Confidence
Cardiomegaly	**62.2%**	cardiac / mediastinal silhouette	Moderate	Moderate

Laboratory Findings & Identified Patterns

Analyte / Test	Result	Unit	Reference Interval	Flag
MCV	**112.4**	fL	80 – 100	**HIGH**
MCH	**36.5**	pg	27 – 33	**HIGH**
RDW	**17.8**	%	11.5 – 14.5	**HIGH**
Hemoglobin	**8.4**	g/dL	13.5 – 17.5	**LOW**
Hematocrit	**25.5**	%	41 – 53	**LOW**
Vitamin B12	**110**	pg/mL	200 – 900	**LOW**
WBC Count	**4.2**	x10^3/uL	4.5 – 11	**LOW**
Platelet Count	**130**	x10^3/uL	150 – 450	**LOW**
Serum Folate	**7.5**	ng/mL	3 – 20	NORMAL

Inferred Diagnoses (ClinicalLabNet Neural Classifier):
• **[MODERATE] Macrocytic / Megaloblastic Anemia** [100.0% Neural Confidence] (Likelihood: HIGH): Neural Network Confidence: 100.0% (ClinicalLabNet deep multi-label classifier). Biomarker pattern is characteristic of macrocytic / megaloblastic anemia. *Pathophysiology:* Deficiency of vitamin B12 or folate disrupting purine/thymidylate synthesis and erythroblast nuclear maturation, leading to nuclear-cytoplasmic dyssynchrony and abnormally large erythrocytes.
• **[CRITICAL] Severe Thrombocytopenia** [86.2% Neural Confidence] (Likelihood: HIGH): Neural Network Confidence: 86.2% (ClinicalLabNet deep multi-label classifier). Biomarker pattern is characteristic of severe thrombocytopenia. *Pathophysiology:* Substantial reduction in circulating platelets predisposing to spontaneous mucocutaneous hemorrhage, petechiae, and visceral bleeding.

Synthesized Clinical Patterns:
• **[SOON] Macrocytic Anaemia:** Low hemoglobin with elevated MCV (>100 fL), suspicious for vitamin B12 or folate deficiency.
• **[SOON] Thrombocytopenia:** Low platelet count increasing risk of mucosal or systemic bleeding.
• **[ROUTINE] Vitamin Deficiencies:** Nutritional deficiency identified in: Vitamin B12.

Suggested Diagnostic Workup

• Serum vitamin B12 and folate levels
• Serum methylmalonic acid and homocysteine
• Peripheral blood smear
• 12-lead electrocardiogram
• Serum NT-proBNP or BNP

Physician Attestation: I, Dr. Alex Vance, hereby confirm that I have reviewed, verified, and adjusted this decision-support synthesis. Decision: **APPROVE**.
Reviewer Notes: No specific notes recorded.
Signature: _____ **Date:** 2026-10-16 09:20 UTC